SE(3)-WAM: What Should a Robot Remember and Predict for Dynamic Manipulation? Supplementary Material

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This supplement documents the measured experiments, implementation, and remaining study protocols. Sections A–D give the data, configurations, and results; E–F specify pending comparisons; G presents failure diagnostics and offline predictions.

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A. Tasks, Data, and Evaluation

A.1. The Reported Fourteen-Task Suite

The completed studies use the migrated DB0 v0.6 demonstration pool and its fourteen-task evaluation suite. Table S1 lists the task identifiers used in the result arrays. The task-balanced score includes every task, including zero-success conditions.

The common training pool has 11,200 episodes, 800 per task. The internal representation screen uses 350 validation episodes; adapted systems retain their own validation settings (including 1,400 for DreamVLA/OpenWAM). The future screen samples 160,416 chunks per epoch. Stored episode count and effective chunk exposure measure different aspects of training. Later fresh-data production retains its own data identity and does not retroactively change the source of these results. Motion-only P0 prediction has no corresponding action-learning score in the fourteen-task table.

Success is read from the native task evaluator after its ordered physical requirements (Table S1). The accepted FUTURE0, BASE2, and HISTORY1 packages reference the same archived evaluator source, task configurations, and fourteen-suite scenario manifest. We recover the definitions from that archive. Success is checked at each 2 ms physics step. Interrupted continuous conditions reset their counters. For T2, the sample completing the low-relative-motion counter also starts the stable-dwell counter; the two durations should not be added as an exact 0.600 s requirement.

Terminal events. The native result mapping groups events into five categories: *invalid* (invalid state); *boundary* (workspace or downstream exit, drop, expired goal window); *safety* (unsafe contact, excessive rim impulse, wire contact, stage block, release impulse, early robot contact, tray or gripper–support contact, striker trap); *task violation* (blocked driver, release outside the goal, path deviation or backtracking, premature release, stopping/deflecting without capture, unstable tipping, missing/extra impact, wrong striker contact, or insufficient outgoing motion change); and *timeout*. These categories describe evaluator outcomes.

There is no global priority among categories. Each family selects the first terminal event in its physics-step evaluation order and preserves that result. Grasp checks invalid state, workspace exit, downstream exit, and drop before success, then timeout. T1 inserts unsafe occluder contact after invalid state. T2 checks invalid state, rim impulse $> 0.10\,\mathrm{N\,s}$, drop, goal-window expiry, and release outside the goal before success and timeout. T3 checks invalid state, stage block, drop, workspace exit, wire contact, premature release, path deviation, and backtracking before success and timeout. Wire failure uses impulse $> 0.020\,\mathrm{N\,s}$ or penetration $> 0.004\,\mathrm{m}$ for more than two consecutive steps; path deviation is $> 0.030\,\mathrm{m}$ for 0.050 s, and backtracking exceeds 0.05 relative to maximum progress.

T4 checks invalid state, release impulse $> 0.002\,\mathrm{N\,s}$, unsafe ramp contact, unsupported tray contact, can tipping, early-contact/boundary branches, and invalid supported capture before success and timeout. A target that has exited the ramp, remains in contact with the downstream support, and has no fingerpad contact at speed $\leq 0.03\,\mathrm{m/s}$ for 0.25 s

Table S1. Native success conditions in the reported fourteen-task suite. Thresholds come from the evaluator source and task configurations bound to the accepted FUTURE0, BASE2, and HISTORY1 packages. A task must pass its preceding stages and avoid terminal failures. All tasks run at 20 Hz with a 2 ms physics step; durations below refer to physical time. COM denotes center of mass.

Task	Ordered success conditions	Limit
grasp; t1_belt, t1_xyz, t1_so3, t1_occ	Approach within 0.10 m; bilateral fingerpad contact for 0.05 s; collision-geometry bottom at least 0.05 m above support; retain contact and clearance for 0.50 s. T1 adds conveyor drive, translation, rotation, or a physical occluder. SO3 has no additional success-angle threshold; unsafe occluder contact terminates OCC.	160 steps 8 s
t2_trans	Approach within 0.10 m. Entire object collision geometry inside the receptacle: planar clearance and floor gap both ≥ -0.002 m, top no higher than the opening. Release requires an opening command < -0.5 and no fingerpad contact. Relative linear/angular speeds ≤ 0.08 m/s and 0.8 rad/s for 0.10 s, followed by a 0.50 s stable dwell.	200 10 s
t2_se3	Approach within 0.05 m and 0.12 rad. Full T-shaped containment: planar clearance ≥ 0.002 m, floor gap in $[-0.002, 0.002]$ m, tilt ≤ 0.05 rad, top below opening. After release, relative speeds ≤ 0.02 m/s and 0.04 rad/s for 0.10 s, then a 0.50 s dwell. The 0.12 rad gate applies to approach.	200 10 s
t3_phase	Start with the native pre-grasp. Phase alignment: distance ≤ 0.090 m and loop-direction error ≤ 0.20 rad. Enter within 0.030 m at progress ≤ 0.15; reach historical maximum progress ≥ 0.60, then ≥ 0.85 and endpoint distance ≤ 0.055 m.	420 21 s
t3_full	Acquire the wand with bilateral contact for 0.05 s, then complete the phase stages above. In a later sample, retain current progress ≥ 0.85 and move at least 0.045 m from both path and endpoint. Finally open (< -0.5) with no fingerpad contact. Releasing before exit fails.	640 32 s
t4_sphere	Approach within 0.10 m; first qualifying contact after ramp exit; any tray contact before success fails. Bilateral hold for nominal 25 ms (12 physics samples, 24 ms). Raise COM by 0.05 m from its height when the bilateral-hold criterion is first met, and retain contact and height for 0.50 s.	120 6 s
t4_sphere_tabletop; t4_slider	Approach within 0.10 m; contact after ramp exit, while supported, in the capture region, and moving at planar speed ≥ 0.08 m/s. Bilateral hold for 0.05 s, COM lift ≥ 0.05 m from the first attainment of that hold criterion, then contact and height for 0.50 s.	120 6 s
t4_can	Same supported-capture sequence. Before first fingerpad contact, a can-axis absolute dot product with vertical > 0.45 fails as unstable tipping. The configuration does not require a transverse closing axis.	120 6 s
t5_replan	Exactly one selected-side impact group completes before robot contact. At 30 ms after the physical impact-end timestamp, branch-signed lateral speed ≥ 0.035 m/s and yaw rate ≥ 0.35 rad/s. Approach within 0.10 m; bilateral hold with grasp-axis error $\leq 18^\circ$ for 0.05 s; COM lift ≥ 0.05 m from the first attainment of that hold criterion and stable hold for 0.50 s. Impact-group lag is 12 ms and maximum contact is 0.10 s.	120 6 s

fails as stopping/deflecting only. T5 checks invalid state, unsafe contact, early robot contact, wrong striker, extra impact/striker trap, insufficient outgoing change, and boundary events before success. At the horizon, zero completed impacts yields missing selected impact; otherwise the result is timeout.

A.2. Scenario Sampling and Replay

The historical E7/agentview recipes specify proposal ranges (Table S2). Continuous min–max fields use uniform draws from NumPy PCG64; discrete choices are equiprobable before physical-feasibility checks. Evaluation loads each accepted scene's complete stored ScenarioSpec and does not resample its parameters. For harmonic translation,

$$p(t) = c + (0, v_y, 0)t + (A_x \sin(2\pi f_x t + \phi), 0, 0),$$

where $\phi \sim U[-\pi, \pi]$. Thus c differs from the initial position by the phase-dependent offset. T4 targets start at zero linear/angular velocity and evolve under free dynamics; no constant target speed is prescribed. Their slope is continuously sampled from $[5, 15]^\circ$. Ramp-local offsets use the ramp center as origin and lateral, downhill, and normal axes: $p_{\text{world}} = c_{\text{ramp}} + R_x(-\alpha)p_{\text{offset}}$. The center is derived from the exit lip and ramp half-size; offsets are not added directly to the lip. T2-SE3 training uses $\omega_z \sim U[-.25, .25]$; validation excludes $(-.08, .08)$. The other thirteen tasks share train/validation proposal recipes.

Occlusion acceptance requires at least eight initially visible frames with visible fraction $\geq .95$, ten to sixty fully occluded frames (fraction $\leq .05$, lasting 0.5–3.0 s), and at least three reappearance frames with fraction $\geq .8$. These are acceptance criteria, not independent sampling variables.

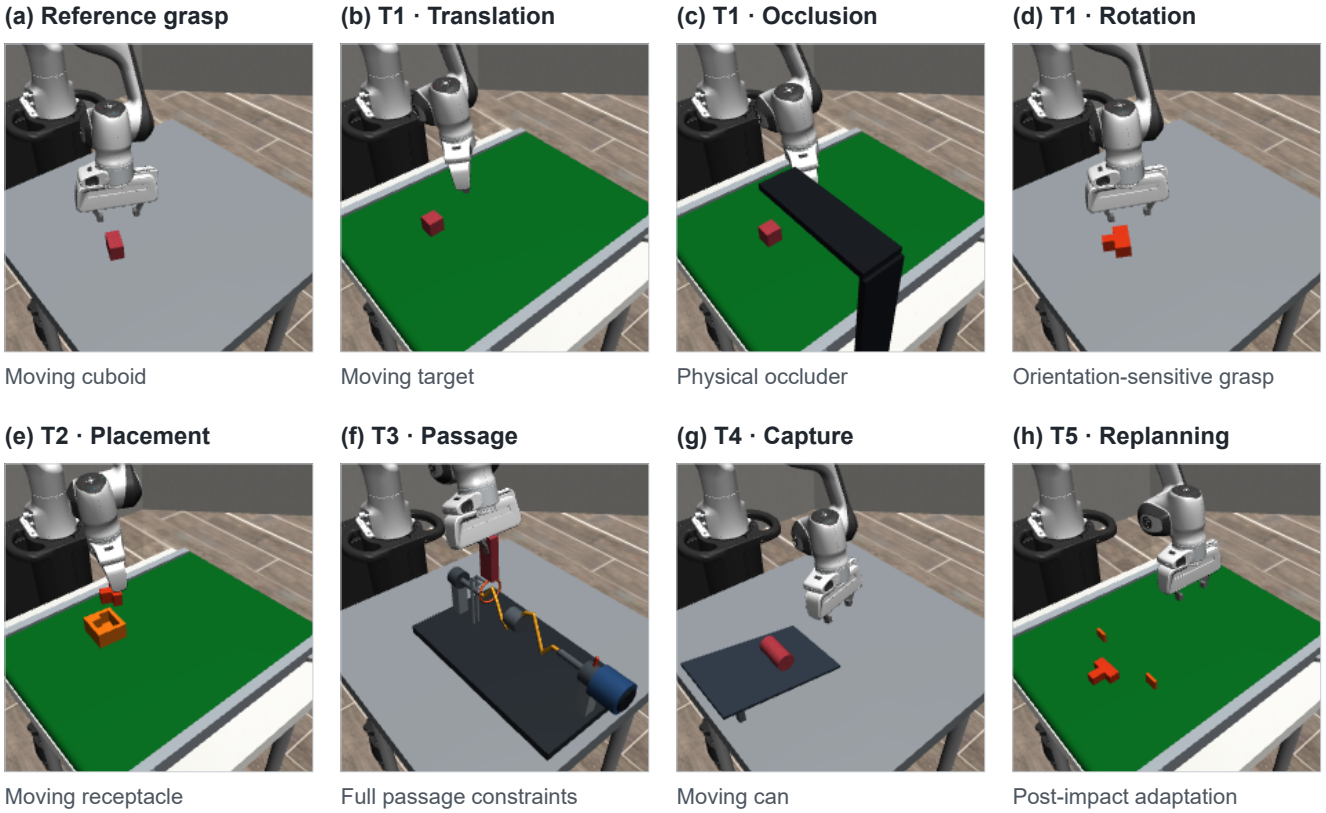


Figure S1. Recorded benchmark observations. Eight examples cover reference grasping; T1 translation, physical occlusion, and orientation-sensitive grasping; T2 moving-receptacle placement; T3 constrained passage; T4 capture; and T5 post-impact adaptation. For each displayed task, the lowest state index in the accepted DEV84 manifest supplies the final observed, uncropped 224×224 agentview RGB. Labels identify task conditions; the snapshots illustrate appearance and geometry, while task outcomes are reported separately. Table S1 lists all fourteen tasks.

For example, the accepted OCC scenes have belt speeds 0.059588–0.099852 m/s within the wider proposal interval. Accepted source seeds can have gaps; a consecutive seed range does not reconstruct the scene set.

All 1,400 stored scenarios share the robot joint initialization

$$q_0 = [0,\ .19634954,\ 0,\ -2.61799388,\ 0,\ 2.94159265,\ .78539816]\ \text{rad},$$

and finger positions $(.020833, -.020833)$ m. T4 additionally stores a temporary preparation-target pose with $x, y \sim U[-.03, .03]$ and yaw $U[0, 2\pi]$, distinct from the release state.

The evaluator source stores recipes in `configs/scenarios/` and task bindings in `configs/tasks/`; the generation entry point is `se3_wam_benchmark/cli/generate_scenarios.py`. The evaluation payload contains the concrete `evaluation_reference/scenes/validation.jsonl` and fourteen `evaluation_reference/scenes/suites/sr_<task>.yaml` files. Each suite uses one repeat and filters the shared manifest by task. These files, with the stored scenario IDs and source seeds, define the replay set used for the paired comparisons.

A.3. Partitions and Statistical Interpretation

The 140-scene development set has ten scenes per task. Its disjoint complement has ninety per task, totaling 1,260. The future screen reports the two partitions separately. The system-reference study also keeps them separate. The combination screen explicitly selects on all 1,400 scenes while retaining both component counts. Thus its selection score includes development data, and the 1,260 complement is now part of the project's development evidence. Neither membership disjointness nor a different rollout seed makes it a final blind test.

The future study uses each condition's own fourth-epoch endpoint and one training seed. Across arms, the scene, source seed, and inference seed are paired. There are 18,900 accepted outcomes, 1,260 per arm; missing scenes and infrastructure-error exclusions are zero. Inference seeds span 20261058–20262317 for the designated selection in-

Table S2. Historical validation proposal ranges. Length, linear velocity, angle, and angular velocity use m, m/s, rad, and rad/s unless marked otherwise. Positions are in world coordinates except the stated ramp-local offsets. Continuous intervals are uniform proposals; discrete choices are equiprobable before feasibility checks. Evaluation replays the stored accepted scenarios, whose distribution need not remain uniform. Harmonic centers are not initial positions.

Task	Initial-state proposal	Motion or event parameters
grasp	Harmonic center $c_x \in [-.08, .08]$, $c_y \in [-.26, -.19]$, $z = .822$; yaw $[-1.1, 1.1]$.	$v_y \in [.025, .125]$, $A_x \in [.015, .035]$, $f_x \in [.3, .7]$ Hz; $\omega_z \in [-1.2, 1.2]$.
t1_belt	Target $x \in [-.08, .08]$, $y \in [-.26, -.18]$, $z = .822$; identity orientation, zero initial twist.	Belt $v_y \in [.04, .10]$.
t1_occ	Same target proposal as Belt; canopy at $(.322, -.29, 1.20)$.	Same belt range; clearance parameter .20; physical occlusion.
t1_xyz	Harmonic center $c_x \in [-.025, .025]$, $c_y \in [-.24, -.20]$, $z = .818$; yaw $[-.35, .35]$.	$v_y \in [.04, .09]$, $A_x \in [0, .03]$, $f_x \in [.25, .65]$ Hz; $\omega_z = 0$.
t1_so3	Same as XYZ.	Same translation; $\omega_z \in [-1.2, -.4] \cup [.4, 1.2]$, equal sign probabilities.
t2_trans	Goal $x \in [-.13, -.08]$, $y \in [-.18, -.12]$, $z = .825$; fixed orientation.	Goal $v_y \in [.03, .06]$; no rotation.
t2_se3	Same goal x, y range, $z = .804$; yaw $[-.3, .3]$.	Goal $v_y \in [.03, .06]$; $\omega_z \in [-.25, -.08] \cup [.08, .25]$, equal sign probabilities.
t3_phase	Wand $x, y \in [-.03, .03]$, $z = .870$, yaw $[0, 2\pi]$; wire phase $[-.25, .35]$.	Wire center $(.10, -.02, .94)$; world-X rotation $\omega_x = -\{2.5, 3, \ldots, 7.5\}^\circ$/s (11 equal-probability values).
t3_full	Wand at $(-.05, -.02, 1.04)$ with identity orientation; same wire phase range.	Same wire rotation; cradle and release requirement enabled.
t4_sphere	Ramp-local offsets: lateral $[-.015, .015]$, longitudinal/normal $(-.145, .030)$; lip $(.06, -.02, 1.10)$.	Gravity-driven from rest; gate release at tick 0; slope $[5, 15]^\circ$; target/ramp sliding friction $[.7, .9]$; unsupported capture.
t4_sphere_tabletop	Lateral $[-.015, .015]$, longitudinal/normal $(-.124, .034)$; lip $(0, -.06, .808)$.	Same slope and release; target/ramp friction $[.7, .9]$, table friction .35.
t4_slider	Lateral $[-.015, .015]$, longitudinal/normal $(-.129, .024)$; supported ramp.	Same slope and release; target/ramp friction $[.025, .045]$, table friction $[.012, .018]$.
t4_can	Lateral $[-.010, .010]$, ramp-axis yaw $[-4, 4]^\circ$; longitudinal position follows gate geometry and oriented can extent.	Same slope and release; target/ramp friction $[.25, .45]$, table friction $[.09, .15]$.
t5_replan	$x \in [-.005, .005]$, $y = -.29$, $z = .812$; yaw $[-.08, .08]$; zero initial twist.	Belt $v_y \in [.12, .14]$; equal-probability left/right striker triggered at $y = -.13$; stroke .098, pulse .35 s, actuator scale $[1, 1.1]$, force cap 4 N; command delay 50 ms.

dices. Matching an inference seed specifies rollout randomness but does not imply that unlike training graphs consumed identical random-number sequences.

For two policies on the same scenes, let b count candidate successes/reference failures and c candidate failures/reference successes. Then

$$\Delta\mathrm{SR} = 100(b - c)/1260 \tag{S1}$$

in percentage points. The report uses the exact two-sided McNemar test on discordant pairs. Fourteen comparisons to P00 and thirteen to P03 form the twenty-seven-comparison family; adjusted values use $\min(1, 27p)$. The 378 task-level comparisons remain exploratory.

We additionally compute 95% pointwise percentile intervals by resampling paired scenes within each task. For every replicate, each task retains ninety pairs (one hundred in the combination study), and the fourteen task differences are averaged with equal weights. Sampling a multinomial from each task's four paired outcome counts is equivalent to resampling its binary scene pairs. We use 100,000 replicates and the 2.5th/97.5th percentiles with linear interpolation. The generator is NumPy 2.3.5 PCG64; seeds are 20261017 plus the zero-based lexicographic comparison index for the twenty-seven future contrasts, and 20261017/20261018 for I versus F/E. The post-hoc I–E analysis excluding phase passage retains thirteen tasks with seed 20261019. Tasks are processed in lexicographic order. These intervals condition on the trained checkpoints and fixed task mixture; they do not include training-seed variation or candidate-selection uncertainty. They are not simultaneous intervals; the corrected

McNemar family above remains the basis for the overall future-screen comparisons.

A.4. Inputs and Time

The completed future screen uses one current fixed-camera RGB view, exact depth/camera geometry, robot-state history, and privileged object information. A simulator field stored beside an image is not automatically a deployable sensor observation. Future target states and label-validity masks supervise their respective branches and are excluded from the action condition. A current-object-pose oracle and a future oracle are distinct information conditions.

Physical high-level steps are separated by 50 ms. All one hundred T5 scenarios use a one-step command delay (50 ms); the other thirteen tasks use zero. The native action queue returns a zero command until the delay elapses, then applies the oldest queued command. Forecast labels cover 50–400 ms relative to the source observation. Synchronous evaluation pauses the environment for policy computation, so it cannot measure the consequence of wall-clock observation age. Later fixed-delay and real-delay evaluations must retain their own time traces and report whether physics continues while the policy computes. Future labels follow the demonstration's recorded geometry under its executed actions. The current validity masks check geometric support, projection, and visibility. They do not establish that the same post-contact future remains valid for an alternative candidate action. Action-consistent label support is **[pending for online generation]**.

B. Reference Implementation and Budgets

B.1. Actual Representation-Screen Configuration

The reference D model uses `se3wam_v2/split_ffn`, four transformer layers, width 768, eight attention heads, and factorized attention. DINOv3 ViT-L/16 is frozen [8]. A 224×224 agent-view image produces 196 current visual tokens. H0V has zero visual-history query tokens. The information track also supplies exact depth and camera geometry. Projected image patches receive a view embedding, a linear embedding of mean valid metric depth, and a linear embedding of the patch-center robot-base XYZ before the view interface. All reported internal conditions use one agent view, so this interface returns that view directly and does not perform cross-view attention. XYZ is backprojected from fixed source pixels (indices 7, 23, …, 215) using camera intrinsics and extrinsics. These coordinates also enter 3-D RoPE for visual Q/K and the corresponding flow queries; object, robot, meta, and action tokens receive no XYZ. Robot input retains eight EEF7 states at 20 Hz. Eight object tokens encode current GT pose and seven motion intervals using 25 numeric statistics and pose/interval/acceleration validity flags.

The 25 object dimensions are $(p, \log R, \Delta p, \log \Delta R, \Delta t, v, \omega, a, \alpha)$, with dimensions $(3, 3, 3, 3, 1, 3, 3, 3, 3)$. The eight frames span -350 to 0 ms. Slots 0–6 encode adjacent intervals at their centers (-325 to -25 ms); only slot 7 retains the current absolute pose. Velocity divides interval displacement by elapsed time, and acceleration differences adjacent velocities at their interval centers. No state preceding the window is used. Each dimension is standardized by training statistics and invalid dimensions are zeroed. Three validity flags are appended before a $28 \to 768 \to 768$ MLP with GELU and a time embedding. Eight object, eight EEF, and 196 visual tokens form the scene bank. Sixteen learned meta tokens are separate inputs. The object-motion values come from simulator truth in this track. The current-pose history track removes historical object poses and all derived interval kinematics while preserving the current pose and robot history. This is a different information condition, not a different name for the same input.

D shares attention QKV/output projections and splits feed-forward networks by world/meta, flow, and action stream. Scene tokens access scene tokens; meta tokens access scene and meta; flow accesses scene and flow; action accesses meta and action. Consequently, flow supervision can change shared weights without an online flow-to-action path. The deterministic pose head pools sixteen final meta tokens.

B.2. Coordinates, Losses, and Generation

Target future translation is cumulative displacement in the fixed metric frame. Rotation is the left-relative vector $\phi_h = \mathrm{Log}(R_{t+\delta_h} R_t^\top)^\vee$. For valid targets $\mathcal{V}$, the pose loss is

$$\mathcal{L}_{\mathrm{pose}} = \frac{1}{|\mathcal{V}|} \sum_{h \in \mathcal{V}} \big[\|\hat{d}_h - d_h\|_2^2 + 0.1\, d_{\mathrm{SO}(3)}(\hat{R}_{t+\delta_h}, R_{t+\delta_h})^2 \big]. \tag{S2}$$

The external weight multiplying this loss is separate from the internal rotation coefficient. A full twist exponential would couple translation to rotation through its Jacobian and is not the implemented parameterization.

The action has eight E7 steps: base-frame translation increment, left-relative rotation vector, and gripper command. Translation and rotation training bounds are 0.05 m and 0.5 rad per axis. The inherited evaluation translation limit is 0.10 m per axis. Training normalization uses the mean and population standard deviation of valid, non-repeated issued E7 commands in the training split, accumulated in FP64 and stored in FP32. Action flow matching uses a unit Gaussian in these coordinates. Each query samples $u \sim U[0, 1)$, shared across its full 8×7 action chunk. Deployment reverses normalization, decodes rotation vectors, and clips physical translation/rotation at 0.10 m/0.5 rad per axis; the gripper command uses its sign. Scene flow uses a metric Gaussian

source with standard deviation 0.02 m. Deterministic auxiliary readouts, including pose, do not consume flow-matching source noise and time.

Only the action generator runs during Q2 deployment, using five Euler steps. Conditioning caching is on; temporal/velocity caching is off. Each replan executes one command. Trainable initialization is fresh for seed 0 rather than inherited from a trained common-reference checkpoint. Global batch is 16. Four epochs produce 40,104 updates and 641,664 chunks. AdamW uses learning rate 2×10^{-4}, betas $(0.9, 0.95)$, $\epsilon = 10^{-8}$, weight decay 0.05, and gradient-norm clipping at 1. After 1,000 warm-up updates, cosine decay spans the four-epoch schedule with a minimum learning-rate ratio of 0.1. Training uses FP32 with TF32 enabled, zero dropout, and an FFN expansion ratio of four. P03 weights action, scene flow, and pose losses by one; Alpha is disabled. For cumulative scene flow F_h, the dynamic weight is $w_h = 1 + 128\,\mathrm{clip}(\|F_h - F_{h-1}\|_2/c, 0, 1)$, where $F_0 = 0$ and $c = 0.014620669186115265$ m. The weighted FM loss divides by three times the valid correspondence count, rather than by the sum of weights. P00, P02, and P05 finish with an eight-device topology; other arms finish with sixteen. Some retained runs have recorded topology transitions. All endpoints retain the declared exposure; topology transitions can change the subsequent numerical training trajectory.

B.3. Alternative Target Definitions

Table S3 specifies the target arrays after omitting batch and query axes. Unless stated otherwise, the eight horizons are 50, 100, ..., 400 ms. Surface correspondence follows fixed source geometry; a future occlusion does not by itself invalidate that identity. Track visibility is a separate label, using depth consistency within $\max(0.001\text{ m}, 0.001z)$ in the camera frame.

Wan receives nine RGB frames at 192×160 normalized to $[-1, 1]$. Its first temporal latent is dropped; the remaining two encode frames 1–4 and 5–8. Codec labels are extracted from actual decoder motion vectors. Target-track sampling starts nearest the mask centroid, then selects the farthest point; row-major order resolves ties. Fewer than 32 points are zero-padded and masked. P08/P09 adapt mechanisms rather than reproduce an entire published model.

Regression objectives average squared error over channels and valid positions. Tracks add mean two-coordinate MSE and visibility binary cross-entropy with logits on valid correspondences. Valid positions are pooled across the batch, rather than averaging episode means. Native pose uses the geometric loss above. Replacement losses P04–P09, native flow, native pose, and action each have outer weight one.

Aalpha adds motion-magnitude MSE with weight 0.5 while preserving P03's Alpha-head initialization. P12, P13, and P14 add future depth, spatial DINO, and dynamic-region RGB supervision, each with weight 0.1. P15 combines the three losses at these same weights. A non-existent head has an inapplicable readout, not a zero error. The combination cannot identify every pairwise interaction. Raw total losses across these objectives are not commensurate.

B.4. System-Reference Training

ACT and Diffusion Policy train from scratch for 50,000 updates at global batch 32. The $\pi_{0.5}$ adaptation starts from `pi05_base` and trains 30,000 updates at global batch 256 [7]. DynamicVLA uses 50,000 updates at global batch 44, initialized from `hzxie/dynamic-vla-DOM` with vision, text, and connector frozen. It uses observations at indices $[-2, 0]$, a 20-step horizon, and normalization fitted on 800 demonstrations. OpenWAM uses a fresh fine-tuning run from the official Alpha foundation, for five epochs, 34,490 updates at global batch 192, with its declared foundation components. DynamicWAM uses its official full initialization and 40,000 updates at global batch 128 [6, 9]. DreamVLA's 20-epoch and 40-epoch schedules use independent training runs, 48,220 and 96,440 updates at global batch 512; the shorter endpoint is not the midpoint checkpoint of the longer schedule [10]. DreamVLA receives seven causal RGB frames, EEF position/Euler orientation, two finger positions, and fixed task text. Its missing wrist view is black; inference uses DDIM10 and classifier-free guidance 1.5. OpenWAM receives current RGB, seven joint values, measured gripper opening, and task text. One agent view and two black wrist views form a 384×320 composite; state is mapped to its native 80-dimensional slots. Ten synchronous joint inference steps produce nine video frames and 32 actions, with video decoding disabled.

Executors also differ: ACT uses temporal aggregation, Diffusion Policy executes eight actions, $\pi_{0.5}$ ten, DynamicWAM sixteen, and OpenWAM thirty-two before replanning. DreamVLA predicts three steps but replans every tick, aggregates the latest three forecasts with exponential weights $\exp(-0.01k)$ over their age order, and executes one command. The internal representation screen executes one. System-level interpretation retains these differences. Matched-budget comparisons must also report each system's inputs, initialization, and execution configuration.

Table S3. Future targets in the representation screen. Shape excludes batch/query axes. Displacements are cumulative from the current state except codec vectors, which retain the decoder's native correspondence. All targets come from the recorded demonstration.

Target	Shape	Support, units, and validity
Scene flow	$196 \times 8 \times 3$	Fixed 14×14 patch centers; base-frame meters; valid source depth/geometry and finite future displacement.
Target pose	8×6	Cumulative translation (m) and left-relative rotvec (rad); valid object pose and query.
P04: Wan	$48 \times 2 \times 12 \times 10$	Frozen Wan2.2 VAE posterior mean and native normalization; future groups end at 200/400 ms; valid query.
P05: dense 2-D	$8 \times 2 \times 224 \times 224$	Fixed source pixels; displacement divided by 224; valid source and finite future projection.
P06: codec	$8 \times 2 \times 64 \times 64$	Each future/current pair is encoded as P/I frames; native vectors divided by 64; actual valid vector regions.
P07: scene tracks	$196 \times 8 \times 3$	Same patch source points; two displacement channels divided by 224 and one visibility logit; valid correspondences.
P08: object DINO	8×1024	Object-mask-area-weighted patch pooling; channel layer normalization; positive mask area.
P09: object tracks	$32 \times 8 \times 3$	Deterministic farthest-point sampling in the current target mask; displacement and visibility as P07; padded points invalid.
A_α: magnitude	196×8	Cumulative flow norm (m); same correspondence mask as flow.
P12: depth	$8 \times 1 \times 224 \times 224$	Metric future depth; finite positive pixels.
P13: DINO	$8 \times 196 \times 1024$	Frozen DINOv3-L/16 patches, channel layer normalization; valid query/horizon/patch.
P14: dynamic RGB	$8 \times 3 \times 224 \times 224$	RGB divided by 255; visible surfaces moving more than 1 mm in world coordinates, including robot and other geometry.

Table S4. Training exposure and development counts for all system-reference endpoints. Batch is global; all endpoints use training seed 0. Initialization and execution are part of each adapted system, not matched factors. The two DreamVLA endpoints are independent training schedules. The 1,260-scene counts appear in the main system table.

System	Updates	Batch	Success /140	Execution / initialization note
ACT	50,000	32	50	Scratch; temporal aggregation
Diffusion Policy	50,000	32	54	Scratch; execute 8 actions
$\pi_{0.5}$	30,000	256	89	`pi05_base`; execute 10
DynamicVLA	50,000	44	24	Official DOM checkpoint; native continuous pipeline
OpenWAM-Alpha	34,490	192	76	Official Alpha foundation; execute 32
DynamicWAM	40,000	128	72	Official full initialization; execute 16
DreamVLA: 20 epochs	48,220	512	112	Independent schedule; aggregate forecasts, execute 1
40 epochs	96,440	512	114	Independent longer schedule
Study P00	40,104	16	88	Fresh task network; frozen DINO-L; execute 1
Study P03	40,104	16	93	Same reference recipe as P00
Study I	40,104	16	96	Fresh CNN condition; full visual history; execute 1

C. History Encoding and Conditional Information

C.1. Completed Conditions

The current-pose study compares eight history encodings plus a fresh GT-history reference. Every visual-history condition preserves current GT target pose and eight robot-state samples, while historical object pose and derived kinematics are masked. Its reference future supervision is held fixed. All eight current-pose conditions are complete: current image, Window, Codec-MV, Flow, Flow12D, PACT-Age, Wan, and PACT-Uniform. The fresh GT-history reference also has complete training, 1,400-scene evaluation, and probe reports. It retains current RGB and adds privileged object-motion history. PACT-Age distributes fifty-six history queries from older to newer frames as $[2, 4, 6, 8, 10, 12, 14]$; PACT-Uniform uses seven groups of eight. These adapt PACT-WAM's temporal allocation principle [5] through per-frame query compression, rather than reproducing its complete model.

All windows contain eight samples at 20 Hz, spanning -350 to 0 ms. Current/Window/PACT add fixed 768-dimensional sinusoidal frame-time embeddings in seconds; time subtraction is performed in FP64 before conversion to

Table S5. History encoders in the current-pose study. All conditions preserve the current 196 DINO-L visual tokens, T8 EEF states, and current object pose. History tokens are width 768 and carry no XYZ. Counts below concern added visual/numeric history tokens, not allocated scene slots.

Condition	Sampled times (ms)	History encoding	Added tokens
Current	0	Past image slots are masked.	0
Window	$-350{:}50{:}0$	Seven past frames, 196 DINO patches each; no query compression.	1,372
Flow	$-350, -250, -150, -50, 0$	Four $2 \times 64 \times 64$ flows; each RGB+validity field yields 64 patches and 14 learned-query outputs.	$56 + 4$
Flow12D	Same as Flow	Same visual encoder; four numeric slots receive twelve motion statistics and their flags.	$56 + 4$
Codec-MV	Same as Flow	Four actual codec MV fields, encoded by the same RGB+validity backend; no numeric slots.	56
Wan	$-350{:}50{:}0$	Repeat current frame once; $48 \times 3 \times 14 \times 14$ VAE latent $\rightarrow$ 147 patches $\rightarrow$ 56 queries.	56
PACT-Uniform	$-350{:}50{:} -50$	Shared eight-query bank reads each of seven past frames.	56
PACT-Age	Same as Uniform	Shared query bank interpolated to $[2, 4, 6, 8, 10, 12, 14]$ per frame.	56

FP32. Window retains every past patch. PACT uses one shared eight-query bank; age allocation linearly interpolates that bank with aligned endpoints, rather than adding frame-specific query parameters. Layer-normalized multi-head attention produces query-residual outputs.

Flow is bidirectional Farnebäck flow [2] on area-resized 64×64 RGB (pyramid scale 0.5, three levels, window 15, three iterations, polynomial neighborhood 5, sigma 1.2, flags 0). The four intervals last 100, 100, 100, and 50 ms. Validity requires finite vectors, in-frame endpoints, valid bilinear reverse sampling, and

$$\|f + b(x+f)\|^2 \leq 0.01(\|f\|^2 + \|b(x+f)\|^2) + 0.5. \quad \text{(S3)}$$

Flow color uses direction as hue, unit saturation, and magnitude divided by the interval's 99th percentile, clipped to $[0, 1]$, as value. Invalid pixels are black with validity zero. The four-channel field enters a $4 \rightarrow 768$ convolution with kernel/stride 8, followed by layer normalization, 64 learned spatial embeddings, and fourteen shared queries with interval-time embeddings. Attention, residual addition, and layer normalization produce fourteen tokens per interval.

For valid flow $f = (d_x, d_y)$, let $r = \|f\|_2$, bars denote pixel means, and q be the 99th percentile of r. Flow12D supplies

$$(\bar{d}_x, \bar{d}_y, \bar{r}, q, \Delta t, \bar{d}_x/\Delta t, \bar{d}_y/\Delta t, \bar{r}/\Delta t, q/\Delta t, a_x, a_y, \|a\|_2).$$

Acceleration differences adjacent mean velocities over interval-center time gaps of 100, 100, and 75 ms; the first interval has no acceleration. All dimensions except Δt are divided by 64, with no fitted mean/standard deviation. Twelve feature-valid flags form a 24-dimensional input to a $24 \rightarrow 768 \rightarrow 768$ GELU MLP. A separate $2 \rightarrow 768 \rightarrow 768$ MLP embeds the left/right times relative to the current frame. Pure Flow zeros the numeric values and flags but retains these four slots and their time embeddings. Thus the comparison adds magnitude, velocity, and acceleration statistics to a color encoding that normalizes magnitude.

Codec labels come from independent MPEG-4 Part 2 I/P pairs with no B frames, GOP 2, quantizer 2, YUV420p, one thread, and bitrate 10^6. Negated decoder backward vectors remain attached to destination blocks, unlike source-grid optical flow. Intra, uncovered, and out-of-frame-reference regions are invalid; no forward/backward confidence filter is applied. Wan uses a frozen FP32 Wan2.2 VAE posterior mean after right-padding T8 with the current frame. Its temporal groups end at $-350, -150, 0$ ms, with real-frame fractions $1, 1, 0.75$. A $48 \rightarrow 768$ kernel/stride-2 convolution creates 147 source tokens; learned spatial embeddings and an MLP for group time/fraction precede 56-query compression. VAE caches are cleared per window.

The original implementation allocates 1,584 scene slots for Current/Window, 272 for Flow/Flow12D, and 268 for the other compressed conditions. These counts include masked past-object slots; sixteen meta tokens are additional. Allocated slots and valid information are reported separately.

Each of the nine conditions has probe extraction over 350 episodes and 1,050 windows with 306 valid readouts. The perturbation analysis below uses the eight current-pose conditions' held-out probe split; the nine-condition raw input/noise/label array audit remains **[pending]**. The completed point estimates give no visual-history gain over the current-image control. The nine full result files match on task, scene, source seed, and inference seed. Against Current, gain/loss counts are 137/223 for Window, 133/148 for Codec-MV, 176/204 for Flow, 183/208 for Flow12D, 132/246 for PACT-Age, 194/232 for Wan, 122/200 for PACT-Uniform, and 238/162 for GT history. The main-paper intervals use Appendix A's task-stratified bootstrap with 100 paired scenes per task and 100,000 replicates. Seeds are

Table S6. Complete history results. The total is a development summary, not a separate held-out policy test.

Condition	/140	/1,260	/1,400	SR (%)
Current image	80	668	748	53.43
Window	69	593	662	47.29
Codec-MV	77	656	733	52.36
Flow	76	644	720	51.43
Flow12D	75	648	723	51.64
PACT-Age	65	569	634	45.29
Wan	70	640	710	50.71
PACT-Uniform	68	602	670	47.86
Reference with privileged target history				
Fresh GT history	84	740	824	58.86

20261017–20261024 in this order. They describe scene uncertainty for the E4 seed-0 endpoints on the development pool, without training-seed, selection, or multiplicity adjustment. This outcome-pairing check does not complete the probe-array audit.

GT history has 586 joint successes and 414 joint failures with Current; its net gain is 76 scenes, or 5.43 points (95% interval 3.07–7.79). Table S7 shows a pronounced task tradeoff: XYZ grasping gains forty successes, while phase passage falls from 84/100 to 1/100. The aggregate score therefore does not imply a uniform gain from motion input.

C.2. Recipe Differences and Required Contrasts

This study uses frozen DINO-L, retains the original Alpha objective, and has a cosine schedule over the first 10,026 updates followed by three epochs at 2×10^{-5}. The combination study uses DINO-S or CNN, disables Alpha, and applies cosine scheduling across all 40,104 updates. Its stronger score therefore cannot be subtracted from the history-study score to estimate a visual-history effect. Flow12D and Wan train fresh on sixteen PRO6000 devices; PACT-Uniform trains on eight PRO6000 devices, with a same-topology restart from update 781. Window, Codec-MV, and Flow resume sixteen-device A100 prefixes on PRO6000; PACT-Age additionally changes from sixteen to eight devices. All preserve global batch 16 and the four-epoch exposure. The device histories preclude a matched throughput comparison. GT history runs on eight PRO6000 devices until update 18,060, then resumes on sixteen with a native RNG reset. Its completed endpoint has 40,104 updates and 641,664 sampled chunks. This topology change is included in the interpretation of the single-training-seed comparison.

The planned core comparison crosses current-pose versus GT-motion input with current-image versus visual-history encoding under a common recipe. Multiple training seeds and paired scenes are needed to estimate the conditional history benefit. A motion probe and an action intervention answer different questions: recoverability of velocity does not demonstrate that the policy used velocity to improve interaction.

C.3. Offline History Perturbations

For each of fourteen tasks, the probe protocol assigns 15/5/5 distinct episodes to training/validation/test, with three windows per episode. Table S8 uses the resulting 70 test episodes and 210 windows, with initial-noise seed 17. This is a probe split within the 350 policy-validation episodes, not a new policy test set. The intervention repeats the current image in all seven past slots or reverses the seven past images. Depth and calibration follow the same image permutation; the current frame, object information, robot history, timestamps, and masks remain fixed. These perturbed histories need not be physically realizable.

Each intervention re-encodes the window without a previous perception cache, rebuilds the condition, and reuses the actual Action/Flow initial-noise arrays. The native reports confirm zero differences for repeated-baseline and identity controls and for every current-image intervention. Separate three-step checks confirm exact condition-cache reconstruction and zero influence of masked old GT states. Future labels are read only after prediction.

All seven history encodings change the issued first action under these perturbations, despite their lack of a success-rate gain in the main screen. For example, Window translation changes by 9.83 mm under repetition and 1.92 mm under reversal; Wan changes by 20.05 and 24.25 mm. The comparison distinguishes sensitivity to history from task benefit. It does not measure a closed-loop intervention or assign a mechanism to the success-rate differences.

This analysis aggregates the original per-task, episode-averaged reports. Their 1,050 sample identities match across the eight conditions, and all forty source JSON reports match their native archives. Those archives omit the formal probe NPZ arrays, so the report checks do not independently recompute prediction arrays or establish cross-condition input/noise equality. The accompanying fixed-ridge readouts decode features at history, meta, and action-input sites; their errors are not the model's target-pose head errors.

C.4. Matched-Current and Occlusion Diagnostics

History substitutions require matching the current target pose, robot state, view, and task condition while changing valid preceding motion. The matching tolerance, group counts, and history-necessary subset acceptance remain **[pending]**. A substitute that changes the current state tests an additional intervention. Occlusion tests must report visibility and duration in physical time, while sampling-gap tests preserve the meaning of displacement by retaining elapsed time. The planned outputs link motion decoding, action change, contact outcome, and unconditional task success.

Table S7. History-study successes, one hundred scenes per task and condition. The first eight conditions retain current GT target pose; GT motion retains current RGB and adds privileged target-motion history. All use one training seed and the fourth-epoch endpoint; the 1,400 scenes are the development pool.

Task	Current	Window	Codec	Flow	Flow12D	PACT-Age	Wan	Uniform	GT motion
Grasp	66	71	62	65	66	62	62	52	88
Belt	87	48	89	99	94	65	88	61	95
Occlusion	72	67	62	74	70	59	87	70	94
SO(3)	42	36	34	38	44	29	37	17	56
XYZ	41	48	54	53	47	55	62	44	81
SE(3) placement	0	0	0	0	0	0	0	0	0
Trans. placement	13	6	15	13	14	12	40	16	10
Full passage	0	0	0	0	0	0	0	1	0
Phase passage	84	6	99	0	0	28	3	67	1
Can	70	96	82	79	92	79	53	68	100
Slider	84	80	70	80	85	59	73	70	78
Sphere	41	55	46	65	76	38	63	45	66
Tabletop sphere	96	97	98	100	100	77	100	96	99
Replan	52	52	22	54	35	71	42	63	56
Total / 1,400	748	662	733	720	723	634	710	670	824

Table S8. Offline sensitivity to past-image perturbations. Mean change in the issued first action, relative to the unchanged input, over 70 probe-test episodes (210 windows, fourteen tasks). Values are equal-task means of episode means. Translation is L2 change in mm; rotation is geodesic change in degrees. These are action changes, not action errors or success gains.

	Repeat current		Reverse past	
Condition	mm	deg	mm	deg
Current image	0.00	0.00	0.00	0.00
Window	9.83	1.28	1.92	0.34
Codec-MV	10.89	1.59	13.13	2.00
Flow	17.80	2.20	15.15	2.51
Flow12D	16.21	2.43	16.85	2.95
PACT-Age	6.07	0.48	2.17	0.25
Wan	20.05	2.70	24.25	3.98
PACT-Uniform	7.50	0.67	0.33	0.04

No-contact failures remain in the success denominator rather than receiving zero contact error.

D. Future Supervision and Combination Selection

D.1. All Fifteen Future Conditions

All conditions in Table S9 have action supervision. Scene-flow coverage includes target regions, so P01 is not a pure context-only condition. P03 versus P01/P02 compares target-pose and scene-flow objectives under the common action path; it does not isolate an exclusive target/context spatial partition. Main-paper Fig. 3 visualizes all fifteen point estimates with task-stratified paired bootstrap intervals, as specified in Appendix A.

P00/P01/P02/P03 test no future objective, scene flow, pose, and their combination. P04–P07 replace the scene target while retaining pose. P08/P09 replace the object target while retaining scene flow. Aalpha and P12–P15 add supervision to P03. Full target definitions and source units are given in Appendix B. None of these conditions changes deployment into online forecast consumption.

D.2. Paired Outcomes and Task Heterogeneity

Table S10 gives gains, losses, difference intervals, and adjusted exact McNemar probabilities against both reference conditions. P05's adjusted probabilities are 0.00844 versus P00 and 0.01877 versus P03. The remaining positive overall differences are not established advantages. The absence of a detected difference is not an equivalence test.

Table S11 gives every task count with equal denominators. The strongest task-specific changes are exploratory. For P05 versus P03, rolling-sphere success falls from 50 to 23 while `tray_contact` failures rise from 40 to 67. Conversely, post-impact replanning rises from 54 to 60. P09's gains in translational placement and replanning coexist with fewer orientation-sensitive grasps. These observations narrow the conditions to investigate, without treating a native failure code as a learned-mechanism explanation.

The lowest count of native safety failures is P06's 250, but its 96 timeouts are the highest. P15 has only 37 timeouts and 343 safety failures. A favorable change in one category must be interpreted alongside all outcomes. Every policy

Table S9. All fifteen future-supervision conditions, grouped by the objective changed. Each uses one training seed, four epochs, common privileged object-motion input, and action-only deployment. The 140 development scenes and 1,260-scene complement remain separate. Scene flow includes target regions.

		Development	Disjoint complement	
ID	Future supervision	Success / 140	Success / 1,260	SR (%)
Reference and component objectives				
P00	No future supervision	88	791	62.78
P01	3-D scene flow	95	782	62.06
P02	Target pose	92	789	62.62
P03	Scene flow + target pose	93	786	62.38
Scene replacements: target-pose objective retained				
P04	Wan latent + pose	92	783	62.14
P05	Dense 2-D displacement + pose	87	746	59.21
P06	Codec motion + pose	86	795	63.10
P07	Sparse scene tracks + pose	90	807	64.05
Target replacements: scene-flow objective retained				
P08	Scene flow + object DINO	89	794	63.02
P09	Scene flow + target tracks	88	808	64.13
Additional objectives: scene flow + target pose retained				
A_α	P03 + motion magnitude	89	778	61.75
P12	P03 + depth	90	791	62.78
P13	P03 + DINO	92	788	62.54
P14	P03 + dynamic RGB	89	790	62.70
P15	P03 + depth / DINO / dynamic RGB	94	802	63.65

Table S10. Paired outcomes over the same 1,260 scenes. Gain means candidate success/reference failure; loss means candidate failure/reference success. Intervals are pointwise task-stratified paired bootstrap intervals for the SR difference, conditional on the trained checkpoints (Appendix A). Exact two-sided McNemar tests use the full family of fourteen P00 and thirteen P03 contrasts: $p_{\mathrm{adj}} = \min(1, 27p)$. P03 against itself is inapplicable.

	P00: no future supervision				P03: scene flow + target pose			
ID	Gain	Loss	95% CI (pp)	Adjusted p	Gain	Loss	95% CI (pp)	Adjusted p
Reference and component objectives								
P01	80	89	$[-2.70, 1.27]$	1	72	76	$[-2.14, 1.51]$	1
P02	61	63	$[-1.83, 1.51]$	1	61	58	$[-1.43, 1.90]$	1
P03	64	69	$[-2.14, 1.35]$	1	—	—	—	—
Scene replacements: target-pose objective retained								
P04	74	82	$[-2.54, 1.27]$	1	71	74	$[-2.06, 1.59]$	1
P05	53	98	$[-5.40, -1.75]$	0.00844	47	87	$[-4.92, -1.43]$	0.0188
P06	81	77	$[-1.59, 2.22]$	1	68	59	$[-1.03, 2.46]$	1
P07	85	69	$[-0.63, 3.17]$	1	81	60	$[-0.16, 3.49]$	1
Target replacements: scene-flow objective retained								
P08	75	72	$[-1.59, 2.06]$	1	59	51	$[-0.95, 2.22]$	1
P09	91	74	$[-0.56, 3.25]$	1	79	57	$[0.00, 3.49]$	1
Additional objectives: scene flow + target pose retained								
A_α	60	73	$[-2.78, 0.71]$	1	53	61	$[-2.30, 1.03]$	1
P12	73	73	$[-1.83, 1.83]$	1	63	58	$[-1.27, 2.06]$	1
P13	70	73	$[-2.06, 1.59]$	1	60	58	$[-1.51, 1.83]$	1
P14	71	72	$[-1.90, 1.75]$	1	62	58	$[-1.35, 2.06]$	1
P15	74	63	$[-0.95, 2.70]$	1	63	47	$[-0.32, 2.94]$	1

Table S11. Future-study successes: ninety scenes per task and 1,260 per condition. Columns follow the task families in Table S1; all tasks are retained, including zero-success tasks. Occl. denotes physical occlusion; Trans. denotes translating-receptacle placement; Tabletop denotes tabletop sphere capture.

	Reference	T1: target motion				T2: placement		T3: passage		T4: capture				T5	
ID	Grasp	Belt	Occl.	SO(3)	XYZ	SE(3)	Trans.	Full	Phase	Can	Slider	Sphere	Tabletop	Replan	Total
Reference and component objectives															
P00	85	90	90	75	76	0	8	3	15	90	69	45	86	59	791
P01	84	90	90	64	69	0	18	0	11	90	74	46	90	56	782
P02	87	90	90	78	73	0	7	0	0	90	71	59	89	55	789
P03	86	90	88	75	77	0	7	0	6	86	80	50	87	54	786
Scene replacements: target-pose objective retained															
P04	83	85	84	73	76	0	20	0	0	90	72	54	89	57	783
P05	84	89	90	66	75	0	10	0	0	90	71	23	88	60	746
P06	86	88	89	79	70	0	6	4	2	89	72	60	85	65	795
P07	82	90	90	78	71	0	17	0	0	89	79	55	90	66	807
Target replacements: scene-flow objective retained															
P08	84	90	90	71	81	0	13	0	0	90	77	60	90	48	794
P09	84	90	89	65	78	0	21	0	0	90	77	59	90	65	808
Additional objectives: scene flow + target pose retained															
A_α	86	90	90	72	75	0	3	0	0	85	73	55	90	59	778
P12	83	90	90	76	76	0	5	0	0	89	77	57	90	58	791
P13	86	90	90	80	73	0	6	0	0	88	78	54	90	53	788
P14	85	89	90	72	79	0	5	0	0	90	80	53	88	59	790
P15	87	90	90	79	81	0	11	0	0	90	75	54	90	55	802

Table S12. Native evaluator failure counts out of 1,260 scenes per condition, in the same design groups as the success tables. Safety is the evaluator category, not a certified safety measure. Every failure remains in the success-rate denominator.

ID	Future supervision	Boundary	Safety	Task viol.	Timeout	Total
Reference and component objectives						
P00	No future supervision	84	297	17	71	469
P01	3-D scene flow	62	324	23	69	478
P02	Target pose	77	331	17	46	471
P03	Scene flow + target pose	76	295	51	52	474
Scene replacements: target-pose objective retained						
P04	Wan latent + pose	79	310	41	47	477
P05	Dense 2-D displacement + pose	87	356	14	57	514
P06	Codec motion + pose	96	250	23	96	465
P07	Sparse scene tracks + pose	47	329	20	57	453
Target replacements: scene-flow objective retained						
P08	Scene flow + object DINO	61	322	42	41	466
P09	Scene flow + target tracks	67	299	35	51	452
Additional objectives: scene flow + target pose retained						
A_α	P03 + motion magnitude	57	315	65	45	482
P12	P03 + depth	62	327	40	40	469
P13	P03 + DINO	71	326	27	48	472
P14	P03 + dynamic RGB	70	322	18	60	470
P15	P03 + depth / DINO / dynamic RGB	59	343	19	37	458

failure remains in the 1,260-scene denominator.

D.3. Combination Selection

The next screen crosses three future choices, two visual encoders, and current-target RGB reconstruction on/off. It preserves a full visual window and privileged target-motion input, disables Alpha and late injection, and trains each condition fresh for four epochs with the full-run cosine schedule. Its twelve candidates and the selected configuration's matched P00 and P03 controls are complete. All fourteen conditions start with sixteen devices and global batch 16. L and N resume on eight devices with batch 2 per device, preserving the optimizer, schedule, normalization, sample cursor, and 40,104-update budget, but resetting destination-rank random states. K resumes on a different GPU type with sixteen devices. M completes training on sixteen devices and moves only its remaining 1,260-scene evaluation to eight. I, E, and F retain sixteen-device training. The full grid thus matches nominal exposure, but its P09 candidates do not share one hardware and random-state history.

CNN and temporal conditioning. The trainable CNN has four convolution–GroupNorm–GELU blocks with channels $3 \to 32 \to 64 \to 128 \to 192$. Each convolution has kernel 4, stride 2, padding 1, and no bias; GroupNorm uses eight groups. The 224×224 image becomes 14×14 patches, followed by token LayerNorm ($\epsilon = 10^{-6}$) and a $192 \to 768$ projection. RGB is divided by 255, bilinearly resized with antialiasing, and normalized by means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$. Geometry embeddings follow Appendix B. The full T8 window retains $8 \times 196 = 1,568$ visual tokens, eight EEF tokens, and eight object-kinematic tokens: 1,584 scene tokens when the full window is available, plus sixteen learned meta tokens. Relative-time embeddings and trunk attention fuse the window; there is no history-query compression or temporal mean pooling.

Current-target reconstruction. The mean of the final sixteen meta tokens enters a $768 \to 256 \to 3,072$ MLP with GELU and final sigmoid, producing $3 \times 32 \times 32$ RGB. Simulator target geometry IDs select the currently visible target and its tight bounding box. The crop is resized to 32×32 (bilinear RGB, nearest-neighbor mask); values outside the mask are zero. Loss is masked L1 with outer weight 0.1, divided by the global count of valid RGB elements. An empty mask contributes zero without dropping the training sample. The reconstruction is a training readout, not an action input. I, E, and F share this path, the same D backbone, and the same training normalization from 1,324,374 valid issued actions in 11,200 training episodes. The 350 validation episodes do not contribute normalization statistics.

Candidate I obtains 935/1,400 by the predefined selection score. Against J, the same P07/CNN condition without reconstruction, the paired counts are 817 joint successes, 118 I-only successes, 79 J-only successes, and 386 joint failures. Its net 39-scene gain is distributed unevenly over tasks, as Table S13 shows. The selected candidate is carried forward for testing across training seeds.

The matched P00 control F retains I's trainable CNN, full visual window, privileged motion input, and current-target masked RGB reconstruction (weight 0.1), with future objectives disabled. Both use fresh seed 0, global batch 16, four epochs, and the full-run cosine schedule. F obtains 79/140 and 731/1,260 successes, totaling 810/1,400. For I versus F, the paired counts are 746 joint successes, 189 I-only successes, 64 F-only successes, and 401 joint failures. The 125-scene net gain is 8.93 percentage points (95% pointwise interval 7.00–10.86). It includes 72 on phase passage and 22 on replanning; translational placement loses seven. These are results on the pool used to select I; the interval does not correct for candidate selection.

The matched P03 control E changes the future objectives to scene flow and target pose. It obtains 87/140 and 774/1,260 successes, totaling 861/1,400. I versus E has 786 joint successes, 149 I-only successes, 75 E-only successes, and 390 joint failures. The net 74-scene gain is 5.29 percentage points (95% pointwise interval 3.50–7.07). Phase passage contributes 72 of these net successes; the remaining thirteen tasks contribute two in total. The aggregate difference is therefore highly task-concentrated within this development pool. A post-hoc calculation excluding phase passage gives 863/1,300 for I and 861/1,300 for E: 0.15 points (interval $[-1.69, 2.00]$). This descriptive sensitivity analysis does not replace the predefined fourteen-task score. Confirmation requires unused scenes and additional training seeds.

D.4. Granularity and Prediction-to-Action Evidence

A future comparison must separate role coverage, variable choice, spatial allocation, and compute. The fifteen-target screen changes target packages, not only allocation at equal budget. A direct fine-target/coarse-context test still requires a compatible uniform reference with the same identity information, horizon support, and declared token/compute budget. Its result is **[pending controlled granularity comparison]**.

Physical readouts should compare target translation and rotation at each horizon; flow and tracks require valid correspondences and visibility. Their association with contact error is a separate analysis. The accepted source contains readout arrays, but this manuscript does not substitute flow-matching validation loss for freely generated physical error. The aligned prediction/contact figure requires **[generated predictions, actual states, and contact events]**. If a future

Table S13. Combination-study successes, grouped by future target and visual encoder. DINO is frozen DINO-S; CNN is trained from scratch. Minus/plus denotes current-target RGB reconstruction off/on; parentheses retain source condition IDs. E and F are the selected configuration's matched scene-flow/pose and no-future controls. Each task has one hundred scenes, including development scenes, and each condition has one training seed. Totals use the predefined 1,400-scene selection pool.

	P15: auxiliary				P07: scene tracks				P09: target tracks				P03	P00
Task	DINO − (B)	DINO + (A)	CNN − (D)	CNN + (C)	DINO − (H)	DINO + (G)	CNN − (J)	CNN + (I)	DINO − (L)	DINO + (K)	CNN − (N)	CNN + (M)	CNN + (E)	CNN + (F)
Reference grasp														
Grasp	90	89	91	92	90	92	87	90	79	85	77	87	85	78
T1: target motion														
Belt	100	100	99	100	100	100	100	97	99	100	90	99	100	100
Occlusion	100	100	99	99	99	100	100	97	96	99	82	98	99	98
SO(3)	86	71	80	88	75	87	69	86	69	62	64	78	78	71
XYZ	58	58	82	75	73	75	89	82	44	56	82	84	78	82
T2: receptacle motion														
SE(3)	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Translation	8	4	6	5	9	1	14	4	4	5	9	14	9	11
T3: constrained passage														
Full	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Phase	0	0	0	0	0	0	28	72	0	0	0	2	0	0
T4: object capture														
Can	99	99	99	98	100	99	99	100	97	99	96	99	99	99
Slider	88	85	88	91	74	83	86	81	84	84	88	88	84	77
Sphere	71	67	67	63	47	40	53	56	68	68	62	64	68	45
Tabletop sphere	100	100	100	99	99	100	95	98	100	100	100	100	99	99
T5: post-impact replanning														
Replan	73	63	64	57	74	72	76	72	80	72	79	57	62	50
Total / 1,400	873	836	875	867	840	849	896	935	820	830	829	870	861	810

oracle is introduced, it must use the same action interface and target times. Any oracle-gap ratio is reported only with a supported, non-negligible denominator.

E. Backbone, Information Flow, and Cross-Checks

E.1. D/E Backbone Comparison

The planned backbone comparison uses the confirmed complete history/future configuration, retaining the reference G_0 program where action does not read predicted future. D uses split feed-forward networks and shared attention projections. The adapted E organizes world and action parameters separately within the current model interface.

The planned two-arm comparison trains fresh seeds 0, 1, and 2, with the same initialization sources, data exposure, optimizer schedule, and endpoint. If the current four-epoch recipe is retained, there are 240,624 updates across six runs and 8,400 evaluation episodes at 1,400 per endpoint. These are planned counts. The active-backbone parameter difference is targeted within 5%; frozen frontend parameters and their runtime costs are recorded separately. Matching exposure or parameter count does not match training FLOPs.

The predeclared practical SR and non-inferiority margins are two percentage points, with a 10% p95 latency benefit considered for cost selection. Intervals must support the corresponding superiority, equivalence, or non-inferiority criterion before that label is used. If three seeds are inconclusive, both arms may be expanded to five under the existing plan. Same-device public-policy timing includes internal perception/action computation and excludes external rendering and oracle access. Selection and uncertainty are **[pending]**.

E.2. Seven Information-Flow Conditions

R0 versus R1 tests auxiliary supervision under action-only deployment. R1's two deployment modes test the cost and effect of generating an unused future from the same checkpoint. R1–R4 examine information access. R2 versus R5 and R3 versus R6 compare synchronous and sequential programs. Sequential second stages train on clean first-stage labels and deploy on first-stage predictions. The corresponding train/test mismatch belongs in the interpretation.

Table S14. Planned Q3 conditions; no measured result is assigned to these rows. F denotes a generated future state and A an action state. Arrows specify information access, independently of whether the future loss is enabled. R1 has two deployments of the same trained checkpoint. Sequential second stages use clean ground truth in training and generated first-stage values at test time.

ID	Future loss	$F \to A$	$A \to F$	Cross-stream training input	Inference order
R0	No	—	—	None	A only
R1	Yes	—	—	None	A only
R1	Yes	—	—	None	Joint F and A
R2	Yes	Yes	—	Synchronous noisy states	Joint F and A
R3	Yes	—	Yes	Synchronous noisy states	Joint F and A
R4	Yes	Yes	Yes	Synchronous noisy states	Joint F and A
R5	Yes	Yes	—	Clean F for second-stage A	F, then A
R6	Yes	—	Yes	Clean A for second-stage F	A, then F

The current generative state is scene flow with shape $[B, S, 196, 8, 3]$; pose, tracks, depth, DINO, and RGB can be deterministic auxiliary readouts. A selected sparse-track or Wan representation requires its own generated-state, normalization, noise, and loss definition before this experiment can run. Replacing it with scene flow would change the question. This interface is **[pending final representation binding]**.

The planned three seeds yield twenty-one trained configurations and twenty-four deployment configurations, including R1's second mode; at 1,400 scenes each, that is 33,600 rollout episodes. Alpha is fixed to zero. Per-modality Euler5 is common, but joint inference uses five combined calls and sequential inference uses five plus five. Actual calls, latency, parameters, training exposure, and valid readouts are reported alongside SR.

E.3. Mechanisms and Final Design

An independently replaceable future interface permits oracle, valid matched-current swap, time-shift, or calibrated-noise diagnostics. Each intervention holds perception, robot state, and action-source noise fixed. Changes in action alone show sensitivity; evidence for useful forecast use also requires task and safety outcomes. Removing a branch at inference and retraining without its loss answer different questions.

After selecting information flow, the study revisits retained future representations and applies the selected flow to the unselected backbone. History choices are revisited when the future target changes. Only the necessary comparisons are repeated; previously valid matched endpoints can be reused. Ranking reversals change the conditional design choice. Multi-seed and scenes not used for selection are required before passing a complete (H, Z, B, G) design to pretraining. The selected configuration, cross-check estimates, and mechanism readouts remain **[pending]**.

F. Pretraining, Cost, and External Evaluation

F.1. Measured Policy Cost and Physical Delay

The future-screen policy call means span 177.071–184.212 ms; p95 spans 178.726–185.133 ms; peak allocated memory spans 1.571–1.617 GiB. These are same-device, fixed-order measurements including internal perception and action computation. Loading, external oracle reads, I/O, rendering, and priming are excluded. Each arm uses three recorded eight-frame sequences, six warm-up calls, and thirty measured calls (ten per sequence). After each reset, seven real calls prime the observation history; the eighth is timed with CUDA synchronization before and after the call. The sequences cover translation, occlusion, and post-impact replanning. Inference is FP32 with TF32 enabled, batch/sequence dimensions $B = S = 1$, and five Euler action steps. NVML identifies the shared device as `NVIDIA RPBZZZ6`, with 102,641,958,912 bytes of memory; CPU affinity is 0–2. Memory is the post-warm-up PyTorch allocation, including resident weights. One measurement round supports this range; differences of a few milliseconds do not establish a robust speed ranking.

For deployed commands, action age is the difference between sensor capture and command application. The planned deployment profile separates capture, availability, preprocessing, inference, queuing, issue, and application. It records hardware, precision, batch size, warm-up, synchronization, and timed sample counts. Fixed-delay simulation tests a controlled lag; wall-clock execution tests the actual combined system. The final same-device profiles, deadline misses, and action-age distributions are **[pending]**.

F.2. Recipe and Robot-Data Base

After the design cross-checks, the study fixes input modalities, history support, future targets/horizons, information flow, and action semantics. The next stage optimizes the existing loss weights and training recipe, and changes depth/width within the chosen architecture for scale compar-

ison. Changing the representation or adding an information path belongs to a new design comparison rather than an optimizer ablation.

DROID supplies real robot demonstrations [3]. Preparation checks sensor timing, calibration, depth, command versus measured motion, gripper semantics, and training-only normalization. It does not provide object pose GT; the target-pose pipeline must be separately validated for any selected pose-based objective. Only targets used by the selected design and necessary ablations need to be produced. The data report includes valid-label masks and rejected-action rates. This stage is **[planned]**.

The formal base is trained on real robot data. Two independent post-training branches start from the same base: one for simulation and one for real-robot tasks. Simulation-from-scratch experiments supply recipe experience and a strong reference, while their weights do not initialize the formal robot-data base or the real-robot branch. Comparisons with and without pretraining share downstream data and exposure, and additionally allow equally budgeted recipe searches. They report the pretraining cost explicitly. Data, model, and compute scale have separate axes.

F.3. Planned External and Real-Robot Evidence

Single-arm DOMINO [1] tests dynamics under a different task/data protocol. Static compatibility uses an explicitly selected native suite such as LIBERO [4]. Real conveyor manipulation evaluates perception, prediction, action, and timing together. The final phase reuses completed pretraining, scaling, and independent adaptation results, then fills uncovered object, scene, motion, and delay conditions. It does not repeat an already completed scaling matrix.

The real-robot protocol must specify objects, speed/delay strata, trial allocation, ordering, calibration, safety termination, and human intervention. External native scores are reported separately from the fourteen-task simulation SR. Final training scale, robot trials, and real-world performance are **[pending]**.

G. Diagnostics and Qualitative Evidence

G.1. Supervision at the Executed Action Position

All 800 historical full-passage demonstrations release only at the final action: chunk position h_7 has release supervision, while the executed position h_0 has none. This identifies an action-position coverage gap.

Tail-window training produces 2/16 successes in diagnostic validation and 1/6 in a selected-scene follow-up. For phase passage, three seeds after 256 updates give 0/96, 12/96, and 26/96; at 1,024 updates all three give 0/96. Six first-action-weighting endpoints total 576 failures in 576 trials, and six GT-role diagnostic endpoints also total zero successes in 576. These local budgets are separate from the fourteen-task study.

A zero action-source deployment intervention changes the three 256-update scores from 0/12/26 to 0/1/46 successes out of 96 each. This post-hoc intervention is exploratory. A reproducibility investigation links some same-seed trajectory branching to one-gray-level RGB changes; fixed full-input/RNG inference and fixed-action replay remain consistent.

G.2. Gripper Execution Diagnostics

The gripper investigation compares sign decoding, a fixed 0.1 dead zone, and nearest-label ternary projection and history-hold mappings. The ternary projection maps to $\{-1, 0, +1\}$ with thresholds at ± 0.5. For the reported fifth-epoch endpoint, the 1,260-scene counts are 698 for sign, 742 for the fixed dead zone, 749 for nearest-label ternary projection, and 741 for history hold. The ternary projection does not show a clear overall improvement over the fixed dead zone; history hold adds failures outside the capture subset. Neither is adopted as the production default in the evidence cited here.

The broader fourth-to-eighth-epoch dead-zone report has not passed strict historical trajectory-consistency checks and is excluded from main-table claims. Decoder effects and representation effects require separate comparisons.

G.3. Offline Target-Pose Predictions

Figure S2 compares deterministic fourth-epoch P15 pose readouts with two kinematic baselines on 84 states (six per task). All eight horizons are valid, yielding 672 state–horizon pairs. Quantiles use linear interpolation with equal state weights; states from one episode may be dependent. Median translation error rises from 1.45 mm at 50 ms to 4.04 mm at 400 ms; the corresponding rotation medians are $0.47°$ and $1.07°$. The cases are selected by mean translation error, independently of task success.

Kinematic baselines. CV and CA use the final two and three observed GT poses, respectively, within the same history available to P15. They do not read future labels or instantaneous simulator velocities. The saved raw positions, xyzw quaternions, and FP64 timestamps verify a 50 ms interval d for all states. The resulting last-interval derivatives agree with the model's unstandardized object-input values. For positions p_{-2}, p_{-1}, p_0, CV predicts $h(p_0 - p_{-1})/d$. CA predicts $hv_0 + h^2 a/2$, where

$$v_0 = \frac{3p_0 - 4p_{-1} + p_{-2}}{2d}, \qquad a = \frac{p_0 - 2p_{-1} + p_{-2}}{d^2}.$$

CV retains the final interval velocity; CA extrapolates it from the -25 ms interval center to the current time.

In the fixed robot-base frame, define $\omega_+ = \mathrm{Log}(R_0 R_{-1}^\top)^\vee/d$ and $\omega_- = \mathrm{Log}(R_{-1} R_{-2}^\top)^\vee/d$. Rota-

tional CV uses $\Delta R(h) = \mathrm{Exp}(h\omega_+^\wedge)$. CA sets $\omega_0 = 1.5\omega_+ - 0.5\omega_-$ and $\alpha = (\omega_+ - \omega_-)/d$, and integrates

$$\dot{\Delta R}(t) = [\omega_0 + t\alpha]_\times \Delta R(t), \qquad \Delta R(0) = I.$$

Exponential midpoint steps are at most 1 ms; halving the step changes any predicted rotation by at most $0.000548°$. There is no smoothing, clipping, or fitted baseline parameter.

Table S15. DEV84 offline prediction errors: 84 states and eight horizons, yielding 672 dependent observations. CV/CA use the last two/three GT poses; P15 also uses the full history and its other observation inputs.

	Translation (mm)		Rotation (°)	
Method	Mean	Median	Mean	Median
P15	3.762	2.771	1.772	0.857
CV	12.943	1.444	9.675	0.175
CA	22.413	0.634	10.114	0.170

Table S15 reports both mean and median errors. P15 has lower means but higher pooled medians than either extrapolator: the baselines combine many small errors with a large-error tail. Using each state's mean over eight horizons, P15 has lower translation error than CV and CA on 42/84 and 37/84 states, and lower rotation error on 27/84 and 34/84. Thus the aggregate means do not establish uniform prediction superiority. The comparison characterizes offline error distributions under GT motion inputs.

G.4. Closed-Loop Alignment

The planned comparison will align observations, predictions, actual futures, action application, and contact in physical time. It will include P05's adverse sphere condition, a sparse-trajectory condition, and a matched future-to-action intervention, recording scene/camera identity, units, visibility, outcomes, and selection rules. Synchronized rollout exports, real-robot and forecast-intervention videos are **[pending]**.

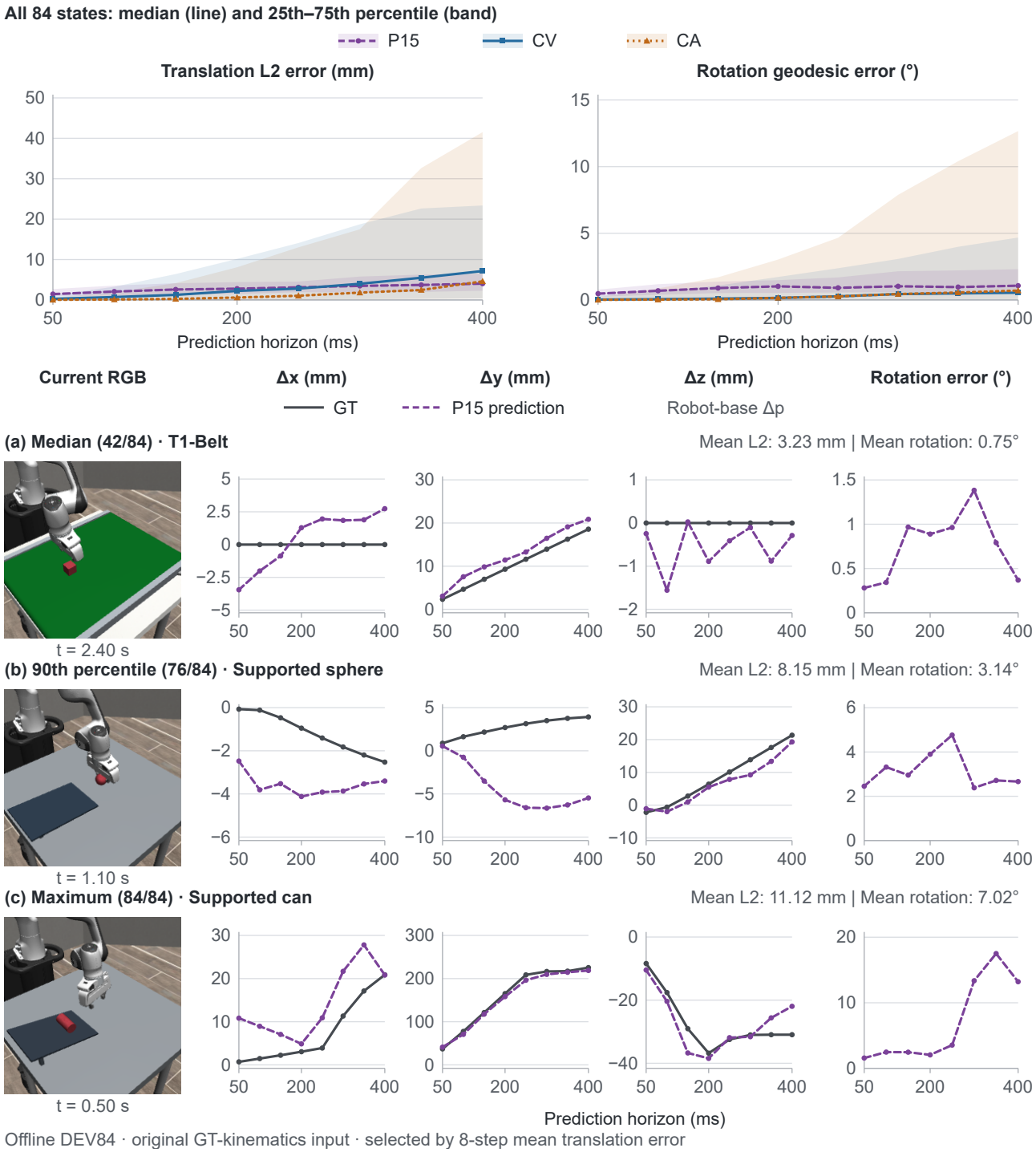


Figure S2. Offline P15 predictions and kinematic extrapolation. The DEV84 probe set contains 84 query states from 28 episodes across fourteen tasks. Top: P15, constant-velocity (CV), and constant-acceleration (CA) median translation L2 and rotation geodesic errors at each horizon; bands show the 25th–75th percentiles across states, not confidence intervals. Bottom: we sort states by their mean translation L2 error over eight horizons and select nearest ranks 42, 76, and 84 (50th, 90th, and 100th percentiles), breaking ties by state index. Each row shows the original current image, predicted and true cumulative translation in robot-base coordinates, and rotation geodesic error at 50–400 ms. The model retains its GT target kinematics and robot-state history. These diagnostic examples characterize offline prediction errors; they do not measure closed-loop contact or task success.

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